

Computer Science Research

- Any research performed using computer science tools and technologies is referred to as computer science research.
- This research area has a long tradition to present the research in the conference publications.
- A computer scientist is expected to write/contribute a good number of peer reviewed papers.
- A student should understand the peer review process.
- Suitable for students: constructive research, exploratory research, case studies, experimental study, design research and so on.

Choosing A Research Problem

- **Supervisor's suggestions:** Supervisor can provide suggestions based on his research area.
- **Suggestions of others:** It is very helpful and useful to read the “future directions” or “future planning” sections of the research papers.
- **Replication of research work with modifications:** This might lead to new findings if the settings of the previous research are changed.
- **Applying of an existing technique in a new domain:** A modified algorithm can be used in an other research field to manipulate data and get a new result.
- **Favorite topic/chapter/field/algorithm read in the courses:** A research problem is chosen based on the best understood topic/chapter/field/algorithm.
- **Challenging of others' findings:** A research effort to disprove the research result of others, if you feel it is not right.

Research Ethics

- Research ethics are the set of ethical guidelines that guides us on how scientific research should be conducted and disseminated.
- Research ethics consists of a core set of **scientific norms**, developed over time and institutionalized in the international research community.
- Research ethics involves the application of fundamental **ethical principles** to research activities which include the design and implementation of research, respect towards society and others, the use of resources and research outputs, **scientific misconduct** and the regulation of research.
- The application of **moral rules and professional codes of conduct** to the collection, analysis, reporting, and publication of information about research subjects, in particular active acceptance of subjects' **right to privacy, confidentiality**, and **in formed consent** (social science).

Avoid Misconduct in Scientific Research

- Fully documentation of results.
- Avoid false presentation of data and results.
- Ensure repeatability and preserve all data.
- Question your own findings.
- No **exaggeration of research claims or outcomes**.
- Acknowledgement of others' work honestly.
- Do not collude in of others.

[Read the ACM and IEEE code of ethics](#)

Research Proposal

- A research proposal is a summary of the proposed research work.
- It addresses the key issues/questions throughout the research study.
- It covers the overall process about how to conduct the research work along with background study and literature review.
- Varies from one field of study to another.
- It clarifies the key objectives and ideas.
- Academics, scientists, researchers generally write research proposal to get fund.
- Students write research proposal to get approval from the supervisor/supervisory committee/institution.

Research Proposal

A research proposal generally includes:

- ✓ **Title:** a tentative title.
- ✓ **Introduction/Background:** introduction, background, problem statement, research objectives & questions.
- ✓ **Literature review:** state and compare related research works, identify strengths and weaknesses, justify how to challenge these works/plan.
- ✓ **Methodology:** states the complete approach and steps to conduct the proposed research; types of research, research methods, tools and technologies to be used, data collection source, experiment.
- ✓ **Expected outcomes/Contribution:** mention the expected contribution of the research work.
- ✓ **Conclusion:** concluding remarks.
- ✓ **Reference list**
- ✓ **Timeline/research schedule/timeframe**
- ✓ **budget**

Reference/Bibliography

- **Cite/Citation:** refers to (a passage, book, or author) as evidence for or justification of an argument or statement, especially in a scholarly work. A way of giving credit to them for their creative works.
- **Reference:** A detailed list of references that are cited in a research work.
- **Bibliography:** A detailed list of references that are cited in a research work and a list of other research related study that are not cited.

Citation Style

- **APA (American Psychological Association):** Generally used by Education, Psychology and Sciences.
- **MLA (Modern Language Association):** Generally used by the Humanities.
- **Chicago:** Generally used by Business, History and Fine Arts.
- **IEEE:** Generally used by Electronics, Engineering, Telecommunications, Computer Science and Information Technology.
- **Sample paper:**
 - Author: Tony Hoare, Robin Milner
 - Article name: Grand Challenges for Computing Research
 - Source: The Computer Journal, Volume 48, Issue 1, 2005, Pages 49–52.

Citation Style

APA Style:

- See: <https://apastyle.apa.org/style-grammar-guidelines/references/examples/journal-article-references#1>
- In-text citation: (Hoare et al., 2005)
- Sample Reference (journal article): Hoare, T, & Milner, R (2005). “Grand challenges for computing research”. The Computer Journal. 48(1), 49-52.

MLA Style:

- See: <https://style.mla.org/>
- In-text citation: (Hoare and Milner 49)
- Sample reference (journal article): Hoare, Tony. “Grand Challenges for Computing Research”. The Computer Journal, vol. 48, no. 1, 2005, pp. 49-52.

Citation Style

Chicago Style:

- See: https://www.chicagomanualofstyle.org/tools_citationguide.html
- In-text citation: (Hoare, and Milner 2005, 51)
- Sample reference (journal article): Tony Hoare, Robin Milner. 2005. “Grand Challenges for Computing Research”. The Computer Journal 48, no. 1: 49-52.

IEEE Style:

- See <https://iee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf>
- In-text citation: refers to the source with a number in a square bracket, e.g. [1], that will then correspond to the full detail in the reference list.
- Sample Reference (journal article): [1] T. Hoare and R. Milner, “Grand challenges for computing research”, Computer Journal, vol. 48, no.1, pp. 49–52, 2005.

Plagiarism

- The practice of using someone else's work/idea and using it as one's own.
 - To steal words/ideas without giving credit to them.
 - Not using of quotation marks.
 - Giving incorrect information of the quotation.
 - Changing words but copying the sentence structure without giving credit.
 - Act of fraud.
- Source: www.plagiarism.org

Possible types of Plagiarism

- Direct Plagiarism: Copying text word-for-word without citation.
- Self-Plagiarism: Reusing your own previous work without permission or acknowledgment.
- Mosaic Plagiarism: Mixing copied phrases or ideas with your own without proper attribution.
- Accidental Plagiarism: Failing to cite sources correctly or paraphrasing poorly, even without intent.

Plagiarism

- Plagiarism should be avoided/prevented:
 - Cite the source properly
 - Use quotation marks
 - Paraphrasing
 - Try to generate own idea
 - Use anti-plagiarism software/plagiarism checker:
 - Grammarly
 - Turnitin

Paraphrasing

- Restatement/rewording of a text/paragraph/information/work/idea in one's own words with proper citation.
- The meaning/idea/work of the original source cannot be changed.
- Alternative to quote (copying someone's exact words and write them in quotation marks with citation).
- How to do paraphrasing?
 - Read and understand the meaning of the sentence/paragraph properly.
 - Write it in your own words (use synonyms, change the structure, break the long and complex sentence into simple one, start from a different point...).
 - Compare your writing with the source so that the original meaning does not get changed.
 - Include citation properly.

Paraphrasing

- Original text: **Cloud computing** is the delivery of computing services such as servers, storage, databases, networking, software, analytics, and artificial intelligence over the internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale. Instead of owning and maintaining physical data centers or servers, users can access technology services on demand from a cloud provider.
- Paraphrased: **Cloud computing** refers to the practice of delivering computing services—like servers, storage, databases, networking, software, and analytics—through the internet. This model enables users to access powerful technology resources from a cloud provider without needing to own or manage physical hardware, paying only for what they use.

Reference List and Readings

- Tony Hoare & Robin Milner. Grand Challenges for Computing Research. The Computer Journal, vol. 48, no. 1, page: 49-52, 2005.
- Serge Demeyer. Research Methods in Computing Science. September, 2011. DOI: 10.1109/ICSM.2011.6080841.
- Jose Nelson Amaral. About Computing Science Research Methodology.
- Buchanan, Elizabeth & Aycok, John & Dexter, Scott & Dittrich, David & Hvizdak, Erin. (2011). Computer Science Security Research And Human Subjects: Emerging Considerations For Research Ethics Boards. Journal of empirical research on human research ethics: JERHRE. 6.71-83. 10.1525/jer.2011.6.2.71.